

Intelligent system for Islamic prayer (salat) posture monitoring

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ABSTRACT

This paper introduced an Intelligent Salat Monitoring and Training System based on machine vision and image processing. In Islam, prayer (i.e. salat) is the second pillar of Islam. It is the most important and fundamental worshipping activity that believers have to perform five times a day. From gestures' perspective, there are predefined human postures that must be performed in a precise manner. There are lots of materials on the internet and social media for training and correction purposes. However, some people do not perform these postures correctly due to being new to salat or even having learned prayers incorrectly. Furthermore, the time spent in each posture has to be balanced. To address these issues, we propose to develop an assistive intelligence framework that guides worshippers to evaluate the correctness of their prayer's postures. Image comparison and pattern matching are used to study the system's effectiveness by using several combining algorithms, such as Euclidean distance, template matching and grey-level correlation, to compare the images of the user and the database. The experiments' results, both correct and incorrect salat performances, are shown via pictures and graph for each of the postures of salat.

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1. INTRODUCTION

Salat (prayer) is one of the main pillars in Islam, which is considered one of the most important aspects of our faith. Our beloved Prophet Muhammad (peace be upon him) received the commandments of salat during Isra' and Mi'raj (the night journey). Hence, giving hope for humanity once more as they have lost the light on how to worship the true and the only one God, The Almighty Allah (Glorious is He and He is Exalted). During that time, Muslims learn how to perform salat by following the orders and actions of Prophet Muhammad (peace be upon him). This is done by looking at the action and orders directly using the senses of sight and hearing. In other words, the technique used to learn salat at that time solely by using the human senses to detect the correct movements and words. Although Muslims during that time can only learn how to perform salat through the Prophet's words and actions, the teaching and learning are highly effective because Muslims nowadays are still performing salat the same way as the Prophet.

As the world is getting older, some Muslims tend to forget the proper way to perform salat as they are bound to the world. Today, technologies are improving at a very fast pace. Many kinds of research and development have been conducted to improve our lives. This raised a big responsibility for Muslim researchers to develop a technology that benefits this world and hereafter. With this goal, developing the